

⇒ TCP/IP Model:

is the practical blueprint of the internet - unlike the theoretical 7-layer OSI model, TCP/IP uses 4 layers and powers real-world networking today.

1: Application Layer:

Function: The layer you directly interact with.

- Protocols: HTTP/HTTPS (web), SMTP (Email), FTP (Files)

2: Transport Layer:

• Analogy: The Logistic company assigning a Packing Number and deciding whether to send the item in one box or split into smaller parcels.

• Function: Manages end-to-end communication, ensures data integrity, and directs traffic to the correct application using port numbers.

• Protocols:

TCP: Reliable delivery with error checking.

UDP: Fast delivery without delivery confirmation. (Like a live broadcast)

3, Internet layer:

- Analogy: Stamping the Destination Street Address and Zip Code on the package to route it to the right city and house.

- Role: Handles logical addressing using IP Addresses and determines the best physical route across networks (Routing).

- Protocols: IPv4, IPv6, ICMP

4, Network Access layer (Link/Physical)

Function: Converts data packets into binary bits (0s and 1s) and transmits them over physical media like Wi-Fi, Ethernet copper cables, or Fiber optics.

Technologies: ethernet, Wi-Fi, Mac Address

OSI vs TCP/IP.

OSI Model (7 layers)

Application, presentation, session

Transport layer

Network

Data link, physical

TCP/IP Model (4 layers)

Application layer

Transport layer

Internet layer

Network Access layer